1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

TLS-like lymphoid aggregates in NMIBC denote a pre-existing organized anti-tumor immune niche that could support BCG efficacy; however, concurrent high ADNP expression in tumor cells coupled with PD-L1 upregulation drives adaptive immune resistance that disrupts TLS-derived immunity, resulting in poorer durable response to intravesical BCG. Consequently, patients whose tumors are TLS-positive but ADNP-high and PD-L1-high will have diminished durable responses to BCG alone and will benefit from combination therapy (BCG + PD-1/PD-L1 blockade or ADNP-pathway targeting). This yields a testable, biomarker-driven strategy: TLS presence plus ADNP-high and PD-L1-high defines a subset for upfront combinatorial approaches to convert non-responders into durable responders.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PD-L1, ADNP
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PD-L1: PD-L1 induction) and that tumor signals are detectable systemically (ADNP mRNA in CTCs and bTMB), yet these diverse biomarker modalities do not always concordantly predict benefit from intravesical BCG.

ADNP: ADNP mRNA in CTCs and bTMB), yet these diverse biomarker modalities do not always concordantly predict benefit from intravesical BCG.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph

Secondary	MSI	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.